UNIT - I

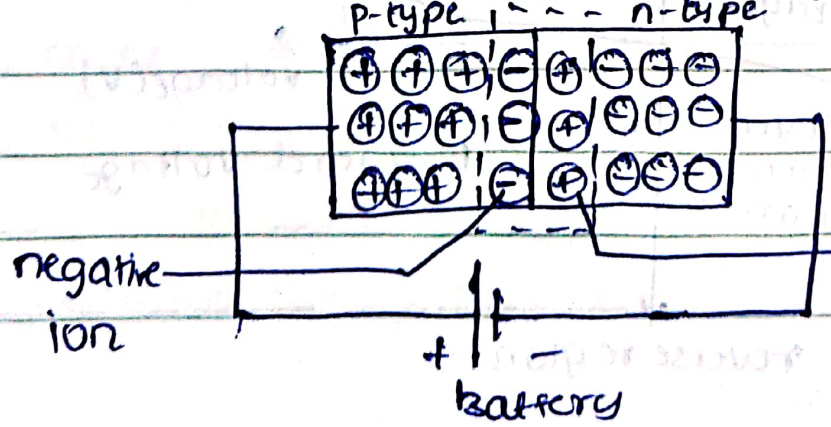
1. Explain the operation of P-N junction diode & draw V-I characteristics:

A. A p-n junction diode is a semi-conductor device formed by joining a p-type semi conductor and an n-type semi-conductor. It allows current to flow only in one direction & blocks it in the opposite direction.

Working of P-N junction diode

(i) Forward Bias:

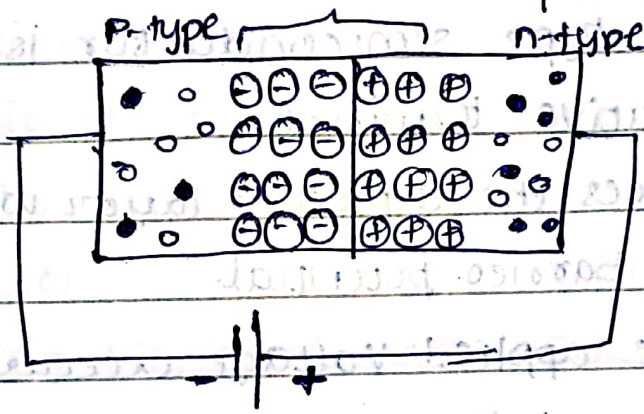
* The p-type semiconductor is connected to the positive terminal of the supply and the n-type semiconductor is connected to the negative terminal.
* This reduces the depletion layer width & lowers the barrier potential
* When the applied voltage exceeds the barrier potential, large no. of carriers cross the junction.
* Hence current flows easily.
* The diode conducts strongly in forward bias



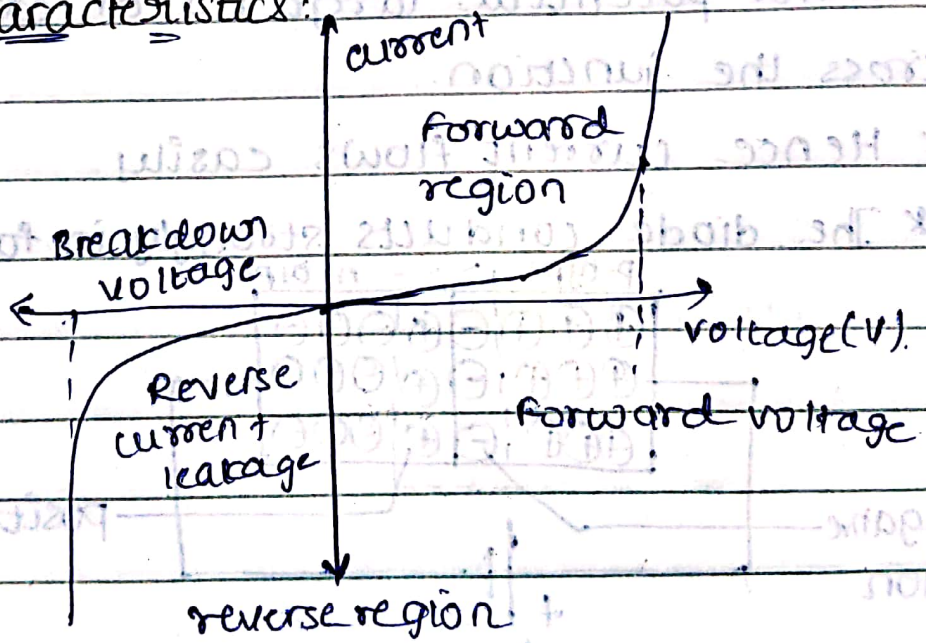
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(ii) Reverse Bias Condition:

* The p-type semiconductor is connected to negative & n-type semiconductor is connected to positive terminal of the supply.
* This increases the depletion layer width and raises the barrier potential.
* Only a very small leakage current flows due to minority carriers.
* The diode does not conduct in reverse bias
* If reverse voltage is increased excessively breakdown occurs. wide depletion region



V-I characteristics:



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Forward bias characteristics:

* Initially, current is almost zero until the applied voltage reaches the knee voltage
* After this point, diode current increases rapidly with a small increase in voltage
* This shows the diode conducts strongly in forward bias.

Reverse bias characteristics:

* A very small reverse saturation current flows due to minority carriers.
* As reverse voltage increases, current remains almost constant until the breakdown voltage is reached.
* Beyond breakdown, current rises sharply.

2. Explain the operation of zener diode & draw V-I characteristics?

A) A zener diode is a specially designed PN junction diode that operates in reverse bias.

→ It is heavily doped so that it can safely enter breakdown region without damage.

Working of zener diode:

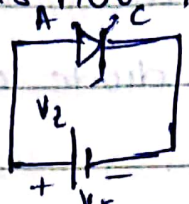
i) Forward characteristics:

* When the p-type is connected to positive

and n-type is connected to negative, the zener diode behaves like normal diode.

* It conducts current after the forward knee voltage (≈ 0.7 Volts) silicon

* Forward region is not mainly used in applications.



(iii) Reverse Bias:

When reverse voltage is applied.

Stge - 1 :- No conduction

* A small leakage current flows.
* Reverse current is negligible
* Diodes blocks current.

This region is called reverse blocking region.

Stage - 2 : Zener breakdown occurs

When reverse voltage reaches a specific value called zener voltage V_z.

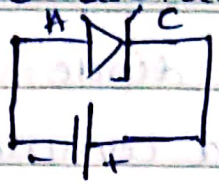
* The depletion layer becomes very thin.
* Electrons tunnel through the junction
* A large current suddenly flows
* Voltage across the diodes becomes almost constant. Which is approxly equals to V_z

This is the zener effect. At higher voltage avalanche effect also occurs.

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Stage-3: Regulation region

* Even if, input voltage increases the voltage across the zener diode remains constant.
* This makes zener diode useful as voltage regulator.



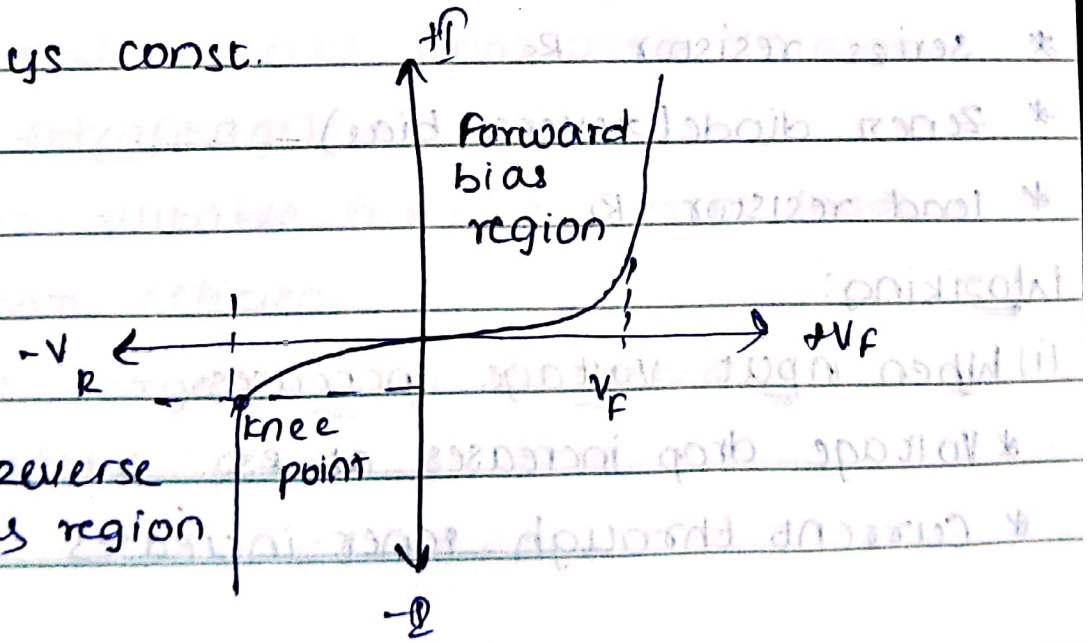
V-I characteristics of zener diode:

(ii) Forward bias:

* Similar to normal diode behaviour
* Small current until the forward knee voltage reached.
* After that, current increases rapidly.

(iii) Reverse bias:

* Initially, only a small reverse saturation current flows.
* When applied voltage reaches zener voltage, breakdown occurs
* Current increases, sharply but voltage stays const.

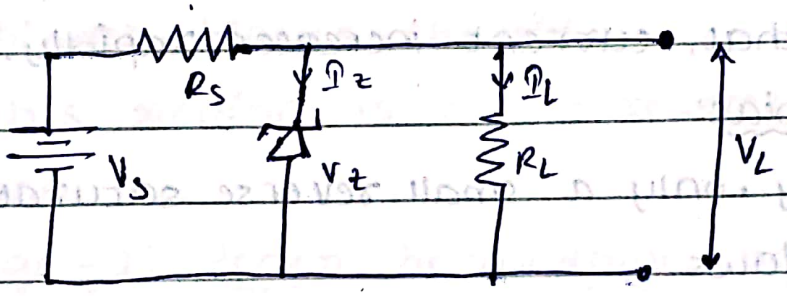


5. Explain how zener diode act as a zener voltage regulator with neat circuit diagram

A. * Zener diode can be used as voltage regulator.

* A zener diode is used in reverse bias to maintain a constant output voltage across a load even when
  (i) input voltage varries
  (ii) load current varies.

This makes it a voltage regulator in power supplies.



Circuit Description:

Components used:

* Input voltage V_n
* series resistor R_S
* Zener diode (reverse bias)
* load resistor R_L

Working:

(i) When input voltage increases:
* Voltage drop increases at R_S
* Current through zener increases

* Zener keeps voltage const at V_z
* Output voltage doesnot change.
* output voltage = V_z

(iii) Input voltage decreases:

* less current flows through R_s
* zener current decreases
* As long as $I_z >= I_z(min)$. V_z remains constant.

(iii) load resistance changes (R_L):

case (i):- load resistance decreases

* More current goes to load.
* zener current reduces.

case (ii):- load resistance increases:

* zener current increases
* voltage across the load still remains V_z.

Advantages:

* Simple & in expensive
* Good regulation for low power
* protects circuit from over voltage

Dis-Advantages:

* Not suitable for high current loads
* Poor efficiency
* At large voltage difference
* Wastes power as heat.

4. Explain the operation of NPN & PNP transistor.

A: (i) NPN Transistor:

In NPN, current carriers are electrons and current flows when the base is made positive

→ Construction:

* NPN transistor has 2 N-type semiconductors & one p-type region.
* Emitter is heavily doped, Based is lightly doped and very thin, collector is moderately doped.

→ Biasing:

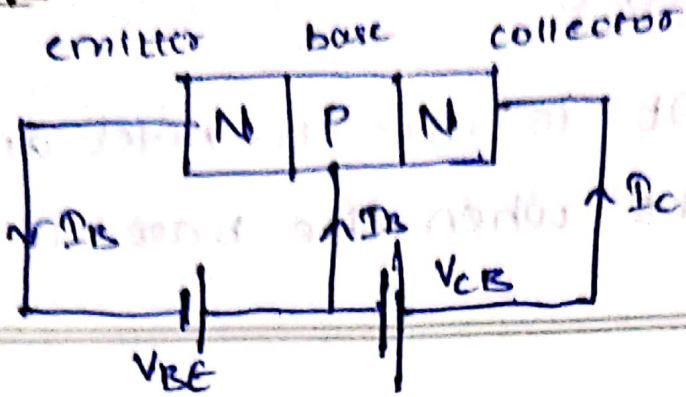
* Emitter - Base Junction (EBJ) is forward biased
* Collector - Base Junction (CBJ) is reverse biased.

→ Working:

* When EBJ is forward biased, ~~holes~~ electrons flow from emitter to base
* only few ~~holes~~ electrons recombine with ~~electrons~~ holes in the base
* Most ~~holes~~ electrons move to collector
* This creates collector current.
* A small base current controls a larger collector current.

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Diagram:-



→ Characteristics:

(a) Input characteristics:

* Graph is similar to a forward-biased diode
* As V_{BE} increases, I_B increases exponentially
* For silicon: cutoff point around 0.7 V.

(b) Output characteristics:

* For a given I_B, the collector current I_C increases with V_{CE} initially.
* After a point, I_C becomes almost const.
* At very low, V_{CE}, transistor enters saturation region.

(c) Transfer characteristics:

* shows relation blw I_C & I_B keeping V_{CE} constant.
* A small change in I_B causes a large change in I_C.
* slope gives current gain 'β'.

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(ii) PNP Transistor:

In PNP, current is due to holes and transistor works when the base is made negative.

→ Construction:

* PNP transistor has 2 p-type regions & 1 n-type region.
* Similar doping levels as NPN.

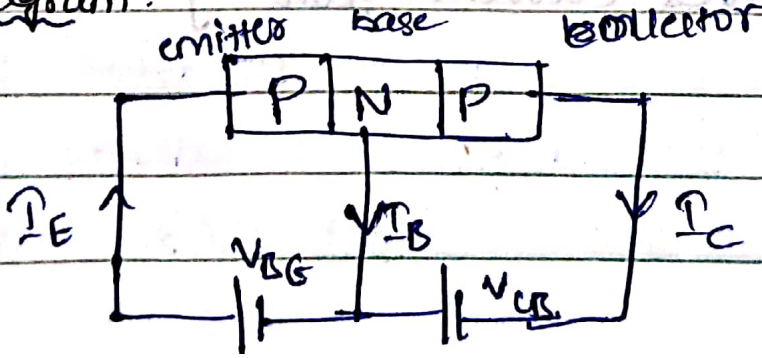
→ ~~Working~~ Biasing:

* Emitter-base junction is forward biased
* collector-base junction is reverse biased

→ Working:

* When EBJ is forward biased, holes flow from emitter to base
* Only few holes recombine with electrons in the base.
* Most holes move to collector.
* This creates collector current
* A small base current controls a larger collector current

→ Diagram:



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→ characteristics:

(a) Input characteristics:

* similar to NPN but voltages are negative
* As V_{BE} become more negative, I_B increases.

(b) Output characteristics:

* shape is same as NPN but lies in the negative voltage region.
* shows cutoff, active and saturation regions.

(c) Transfer characteristics:

* shows linear relation $I_C = \beta I_B$
* β for PNP is usually slightly lower compared to NPN.

5. Explain common emitter mode of transistor & draw input & output characteristics.

A. In common emitter mode, the emitter terminal is common to bot input & output circuits.

* Input is applied b/w base & emitter.
* Output is taken b/w collector & emitter.

This is most widely used transistor because it gives:

* High voltage gain
* High current gain
* High power gain
* High output impedance

Operation:

(i) Input section:

* Made forward biased
* Small input voltage V_{BE} produces a small base current I_B.

(ii) Output section:

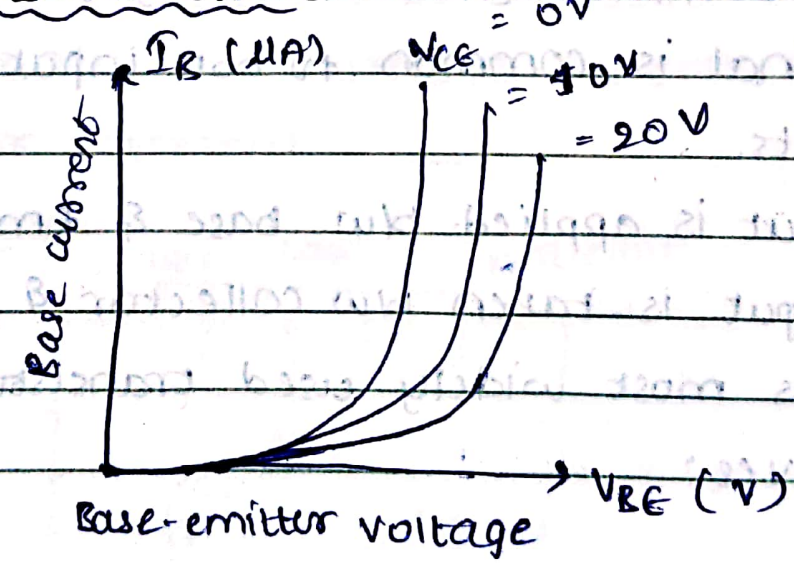
* Made reverse biased
* A large collector current I_C flows, controlled by I_B.

→ Amplification principle:

A small change in I_B causes a large change in I_C.

* The CE mode works excellently as an amplifier.

Input characteristics:



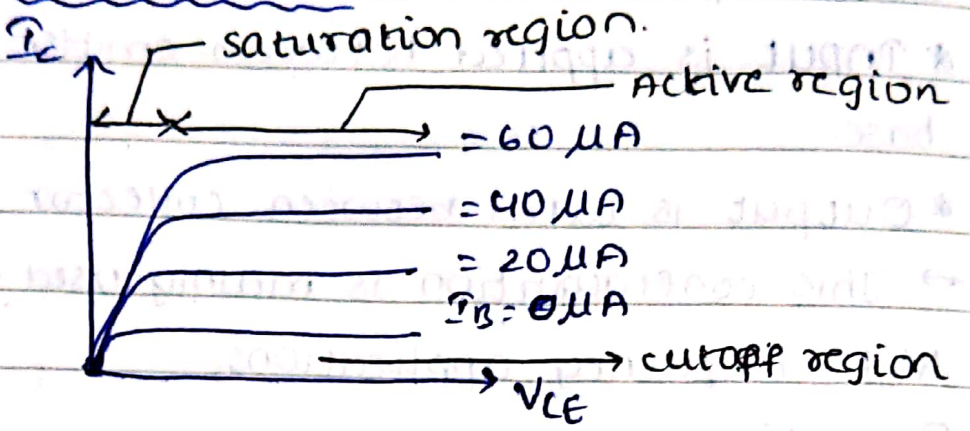
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* At first, I_B is almost zero until V_{BE} reaches ~0.7V for Si

* After that I_B increases rapidly.

* The input impedence is low.

Output characteristics:



(a) Cut-off Region:

* $I_B = 0$
* $I_C = 0$
* Transistor is OFF & both junctions are reverse biased.

(b) Active region:

* I_C increases only slightly with V_{CE}
* I_C depends on I_B
* used for Amplification

(c) Saturation Region:

* At low V_{CE}
* Both junctions forward biased
* I_C reaches maximum.

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6. Explain common base mode of transistor & draw input & output characteristics

A- In the common base configuration, the base terminal is common to both I/p and output circuits.

* Input is applied between emitter and base
* Output is taken between collector & base

→ This configuration is mainly used for high frequency applications.

Operation:

(i) Input Junction:

* Forward biased
* A small increase in emitter voltage V_{EB} produces a significant emitter current I_E

(ii) Output Junction:

* Reverse biased
* A large portion of emitter current flows to the collector as I_C

→ In CB mode,

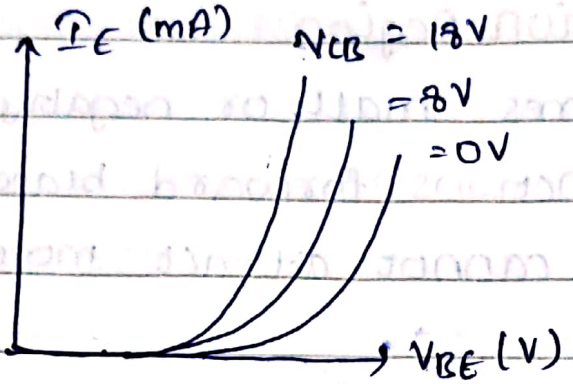
$$\alpha = \frac{I_C}{I_E}$$

i.e. almost the entire emitter current reaches collector

It gives :

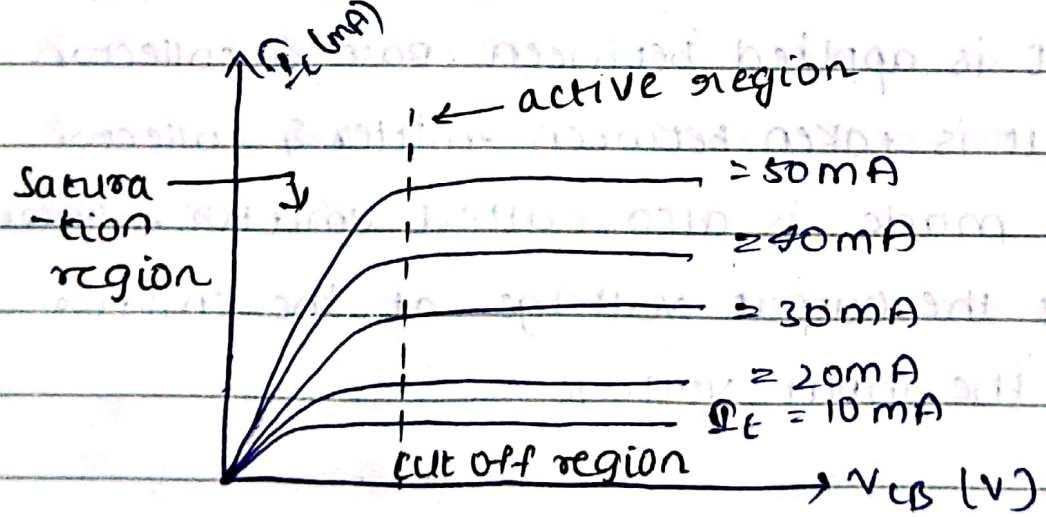
* Very high voltage gain
* current gain is less than 1.
* High output impedence
* low input impedance.

Input characteristics:



* The curve resembles a forward-biased diode.
* I_E increases sharply after $V_{EB} \simeq 0.7V$.
* Input impedance is very low.
* A small increase in V_{EB} causes large increase in I_E.

Output characteristics:



(a) Active region:

* $I_C = \alpha I_E$
* Used for amplification.

(b) Cutoff region:

* $I_E = 0$ & both junctions are reverse biased
* I_C = small leakage current.

(c) Saturation Region:

* V_{CB} becomes small or negative
* Both junctions forward biased
* collector cannot attract more carriers.

3. Explain common collector mode of transistor & draw its input & output characteristics.

A. In common collector mode, the collector terminal is common to both input & output circuits.

* Input is applied between Base & collector
* Output is taken between emitter & collector

→ This mode is also called emitter follower because the output voltage at the emitter follows the input voltage

Biasing:

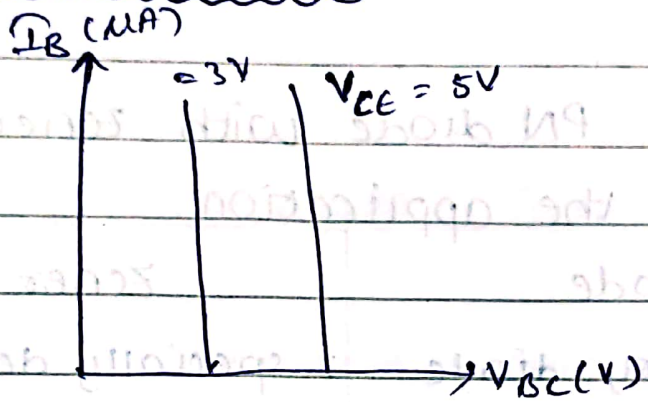
* Base - Emitter junction is forward biased
* collector - Base junction is reverse biased

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Working:

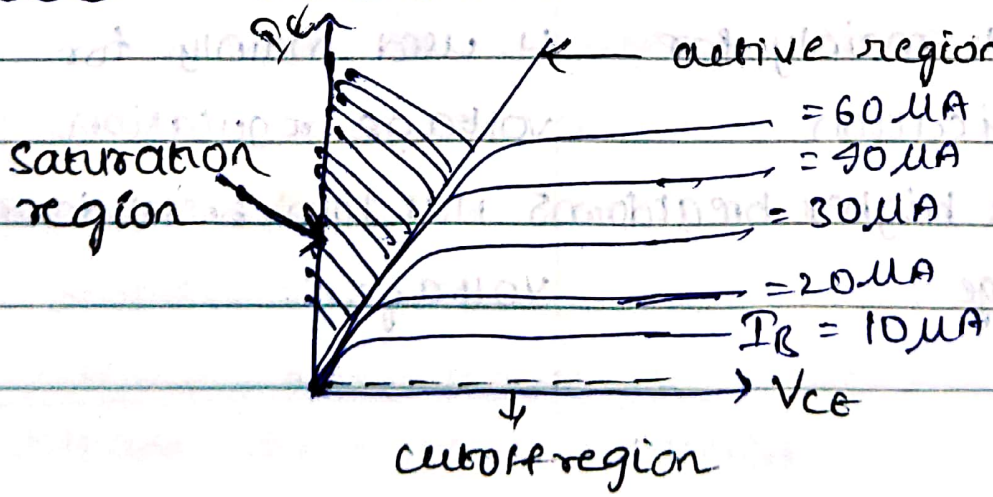
* A small base current I_B controls a large emitter current I_E.
* Output voltage at the emitter is almost equal to input voltage - V_{BE}.
* This makes CC mode excellent for impedance matching.
* Input impedance is very high.
* Output impedance is very low.

Input characteristics:



* Input curve resembles a forward-biased diode
* As V_{BC} increases base current I_B increases.
* Very small voltage gives current.

Output characteristics:



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(a) Active Region:

* CE junction reverse biased
* I_C increases with I_B and is nearly const.

(b) Cutoff region:

* $I_B = 0$ & I_C is almost '0'
* Transistor OFF

(c) Saturation region:

* At V_{EC} very small
* Both junctions forward biased
* I_C cannot increase further.

8. Compare PN diode with zener diode & write the application.

PN diode	Zener diode
1. ordinary diode	1. specially designed diode
2. Allows current only in forward bias	2. Allows current in reverse bias
3. Reverse breakdown is not controlled	3. Reverse breakdown occers at a fixed voltage
4. Used mainly for rectification	4. used mainly for voltage regulation.
5. Has higher breakdown voltage	5. Has low breakdown voltage

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6. cannot operate in breakdown region. 7. No precise voltage control 8. 	6. can safely operate in breakdown region. 7. Provides stable & constant voltage. 9. 

Applications:

(a) PN diode:

* Used in half-wave & full-wave rectifiers.
* Clipping & clamping circuits.
* works as electronic switch in digital logic
* Used in AM radio detectors.
* Used as free wheeling diode in inductive loads.

(b) Zener diode:

* It is used as voltage regulator
* provides a stable reference voltage
* protects sensitive components.
* protects voltmeters & ammeters from high voltage
* used in switching circuits where a specific voltage threshold is required.

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